

Module 3 — Core Skills Drill: SQL Fundamentals

Write three SQL query functions in Python against a provided SQLite database. This drill targets retrieval practice — write the queries from memory, not by copying from the reading.

Due: Tonight (Day 1, after the concept session). Resubmissions are accepted through Saturday of the assignment week.

Setup

1. **Accept the assignment:** [Accept the M3 Drill assignment](#)
2. Clone your repo and open it in VS Code.
3. The repo contains `drill.db` — a pre-seeded SQLite database with three tables:
 - `departments` (`dept_id`, `name`, `budget`)
 - `employees` (`emp_id`, `name`, `dept_id`, `hire_date`, `salary`, `title`)
 - `projects` (`project_id`, `name`, `dept_id`, `start_date`, `end_date`, `budget`)
 - `project_assignments` (`assignment_id`, `emp_id`, `project_id`, `hours_allocated`, `role`)
4. Install dependencies: `pip install -r requirements.txt`

Tasks

Create a file called `drill_queries.py` with three functions. Each function connects to the SQLite database, runs a query, and returns the result.

Task 1 — Aggregation: `top_departments`

Write a function `top_departments(db_path)` that:

- Connects to the SQLite database at `db_path`
- Queries `employees` and `departments` to find the **top 3 departments by total salary expenditure**
- Returns a list of tuples `[(dept_name, total_salary), ...]` sorted descending by total salary

SQL concepts: JOIN, GROUP BY, SUM, ORDER BY, LIMIT

Task 2 — JOIN: `employees_with_projects`

Write a function `employees_with_projects(db_path)` that:

- Returns a list of tuples `[(employee_name, project_name), ...]` for all employees assigned to at least one project
- Traverses: `employees` → `project_assignments` → `projects`

SQL concepts: INNER JOIN across three tables

Task 3 — Window Function: `salary_rank_by_department`

Write a function `salary_rank_by_department(db_path)` that:

- Returns a list of tuples `[(employee_name, dept_name, salary, rank), ...]`
- Rank is computed by `RANK() OVER(PARTITION BY department_id ORDER BY salary DESC)`
- Returns all rows, ordered by department name then rank

SQL concepts: Window function (RANK), PARTITION BY, ORDER BY

Note: SQLite supports window functions (added in version 3.25). Your Python 3.11 installation includes a compatible SQLite version.

Submit

1. Create branch `drill-3-sql-fundamentals`
2. Add `drill_queries.py`
3. Commit, push, and open a PR to `main`
4. The autograder will run and report pass/fail
5. Paste your PR URL into TalentLMS → Module 3 → Core Skills Drill